

Air Quality Index(AQI) Prediction Using Time Series Analysis

(Kolkata, 2017–2025)

A Project Report Submitted in Partial Fulfillment of the Requirements for the
Award of the Degree of

Master of Science in Statistics and Computing

Submitted By

ROSHAN KUMAR
(24419STC024)

M.S.C. Student



**DST-Centre for Interdisciplinary Mathematical
Sciences**

**Institute of Science
Banaras Hindu University
Varanasi – 221005**

2026

Abstract

Air pollution is one of the most critical environmental challenges facing urban India today. The Air Quality Index (AQI) serves as a key indicator of ambient air quality and public health risk. This project presents a comprehensive time series analysis and forecasting system for AQI prediction in Kolkata, one of India's most densely populated metropolitan cities, using daily AQI data spanning the years **2017 to 2025**.

The dataset consisted of daily city-level AQI observations collected across nine years (3,287 records), with approximately 23% missing values (757 entries). The year 2017 was included in the training set; although early 2017 data contained missing values, these were handled through a robust three-stage imputation pipeline: (1) linear interpolation within each Year-Month group, (2) monthly median imputation across all years, and (3) a global median fallback. No outlier capping was applied in this implementation, preserving the full dynamic range of the AQI signal.

The data was aggregated to monthly averages and converted into a formal time series structure of 108 monthly observations. Three time series models were developed and compared: ARIMA (AutoRegressive Integrated Moving Average), SARIMA (Seasonal ARIMA), and Facebook Prophet. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R^2 score on a held-out test set covering January 2024 to December 2025 (24 months).

Prophet achieved the best performance with an R^2 of **0.8308** and MAE of **17.88**, demonstrating its superior ability to capture strong yearly seasonality. SARIMA also performed well with R^2 of 0.7373, while ARIMA achieved 0.5161.

Key findings reveal that AQI in Kolkata is highest during winter months (November to January) due to temperature inversions and low wind speeds, and lowest during the monsoon season (June to September). A clear declining trend in yearly average AQI from 2017 to 2025 indicates improving air quality. A 6-month future forecast (January–June 2026) was generated using SARIMA retrained on the full dataset, with forecast values ranging from 234.44 (January) to 51.25 (June), consistent with the established seasonal cycle.

Keywords: Air Quality Index, Time Series Analysis, ARIMA, SARIMA, Prophet, Kolkata, Forecasting, Missing Value Imputation, Seasonality, CPCB.

Candidate's Declaration

I hereby certify that the work, which is being presented in the project report, entitled "*Air Quality Index (AQI) Prediction Using Time Series Analysis (Kolkata, 2017–2025)*", in partial fulfillment of the requirement for the award of the Degree of Master of Science in Statistics and Computing and submitted to the DST-Centre for Interdisciplinary Mathematical Sciences, Institute of Science, Banaras Hindu University, Varanasi, is an authentic record of my own work carried out during the academic year 2024–2026 under the supervision of **Dr. Rakesh Ranjan**.

I also cited the references about the text(s)/figure(s)/table(s)/equation(s) from where they have been taken. The matter presented in this report has not been submitted elsewhere for the award of any other degree or diploma from any institution.

Date: _____

Signature of the Candidate

Name: Roshan Kumar

Roll no: 24419STC024

Enrollment no: 436261

काशी हिन्दू
विश्वविद्यालय



BANARAS HINDU
UNIVERSITY

Certificate

This is to certify that the project entitled '**Air Quality Index (AQI) Prediction Using Time Series Analysis (Kolkata, 2017–2025)**' has been carried by **Roshan Kumar**, a student of Msc. 4th Semester Statistics and Computing in partial fulfillment for the award of the degree of **DST-Centre for Interdisciplinary Mathematical Sciences** to Banaras Hindu University is a record of original work carried out under my supervision.

The project has fulfilled all requirements according to the academic regulations of the university.

Sign. of Project Supervisor

Dr. Rakesh Ranjan

DST-Centre for Interdisciplinary

Mathematical Sciences

Banaras Hindu University

Signature of Coordinator

Acknowledgements

I would like to express my sincere gratitude to all those who supported me throughout the completion of this project.

First and foremost, I am deeply thankful to my project supervisor, **Dr. Rakesh Ranjan**, DST-Centre for Interdisciplinary Mathematical Sciences, Institute of Science, Banaras Hindu University, for their invaluable guidance, consistent encouragement, and constructive feedback at every stage of this work. Their expertise in data science and time series analysis was instrumental in shaping the direction of this project.

The dataset used in this project was provided by the project supervisor in CSV (Comma Separated Values) format. It contains daily Air Quality Index (AQI) observations for Kolkata city from January 2017 to December 2025. The data was collected from the Central Pollution Control Board (CPCB), India.

I extend my heartfelt thanks to the Head of DST-Center of Interdisciplinary Mathematical Science, Institute of Science, BHU, for providing the necessary academic infrastructure and resources required for this project.

I am also grateful to the **Central Pollution Control Board (CPCB)** and the open data portals for making the Kolkata AQI dataset publicly accessible, without which this research would not have been possible.

A special thank you to my classmates and fellow MSc students for their moral support, technical discussions, and collaborative spirit throughout the academic program.

Finally, I am profoundly grateful to my family for their unwavering love, patience, and encouragement throughout my academic journey.

Roshan Kumar
DST-Centre for Interdisciplinary Mathematical Sciences
Statistics and Computing
Institute of Science, BHU
Varanasi, 2026

Contents

Abstract	1
Candidate's Declaration	2
Acknowledgements	4
List of Figures	7
List of Tables	8
1 Introduction	8
1.1 Background and Motivation	8
1.2 Problem Statement	9
1.3 Objectives	9
1.4 Scope of the Study	10
2 Literature Review	11
2.1 Traditional Statistical Methods	11
2.2 Machine Learning Approaches	11
2.3 Prophet Model	12
2.4 AQI Studies in Indian Cities	12
2.5 Research Gap	12
3 Proposed Approach	13
3.1 Overall Methodology	13
3.1.1 Data Preprocessing	14
3.1.2 Missing Value Handling	14
3.2 Time Series Preparation	15
3.3 Model Selection	16
3.3.1 AutoRegressive Integrated Moving Average Model	16
3.3.2 Seasonal AutoRegressive Integrated Moving Average Model	17
3.3.3 Prophet Model	17
3.4 Evaluation Metrics	18
3.4.1 Mean Absolute Error	18

3.4.2	Mean Squared Error	18
3.4.3	Root Mean Squared Error	19
3.4.4	R^2 Score	19
4	Results and Discussion	20
4.1	Exploratory Data Analysis	20
4.1.1	Dataset Overview	20
4.1.2	AQI Distribution	20
4.1.3	Seasonal Patterns	21
4.1.4	Yearly Trend	22
4.1.5	Monthly Time Series	23
4.1.6	Stationarity and ACF/PACF Analysis	23
4.1.7	Train-Test Split	24
4.2	Model Performance	24
4.2.1	ARIMA(2,1,2) Forecast	24
4.2.2	SARIMA(1,1,1)(1,1,1)[12] Forecast	25
4.2.3	Prophet Forecast	25
4.2.4	Prophet Component Decomposition	25
4.2.5	All-Models Comparison	26
4.2.6	Quantitative Performance Comparison	27
4.2.7	Discussion of Results	27
4.3	6-Month Future Forecast (January–June 2026)	28
5	Conclusion and Future Work	29
5.1	Conclusion	29
5.2	Limitations	30
5.3	Future Work	30
6	Implementation	31
6.1	Tools and Technologies	31
6.2	Data Loading and Combination	31
6.3	Missing Value Treatment	32
6.4	Time Series Preparation	33
6.5	Stationarity Test (ADF)	33
6.6	Train-Test Split	33
6.7	Model Training and Forecasting	34
6.7.1	ARIMA Training	34
6.7.2	SARIMA Training	34
6.7.3	Prophet Training	34

A Python Environment and Dependencies	37
B Dataset Description	38
C Complete Project Notebook Structure	39

List of Figures

4.1 Distribution of Daily AQI Values – Kolkata (2017–2025)	21
4.2 Average AQI by Month – Kolkata (2017–2025)	22
4.3 Yearly Average AQI Trend – Kolkata (2017–2025)	22
4.4 Monthly Average AQI Time Series – Kolkata (2017–2025)	23
4.5 ACF and PACF Plots of the Monthly AQI Series	23
4.6 Train-Test Split of the Monthly AQI Time Series	24
4.7 ARIMA(2,1,2) – Actual vs. Predicted AQI (2024–2025)	24
4.8 SARIMA(1,1,1)(1,1,1)[12] – Actual vs. Predicted AQI (2024–2025) . .	25
4.9 Prophet – Actual vs. Predicted AQI (2024–2025)	25
4.10 Prophet Model Component Plots: Trend and Yearly Seasonality . . .	26
4.11 AQI Forecast – All Three Models vs. Actual (2024–2025)	26
4.12 R ² Score Comparison Across All Three Models	27
4.13 SARIMA – Next 6 Months AQI Forecast (January–June 2026)	28

List of Tables

1.1 India AQI Categories and Health Implications	8
4.1 Average Monthly AQI – Kolkata (2017–2025)	21
4.2 Model Performance Comparison on Test Set (January 2024 – December 2025)	27
4.3 6-Month AQI Forecast – Kolkata (January–June 2026), SARIMA Model	28
6.1 Libraries and Their Roles	31
B.1 Dataset Summary	38

CHAPTER 1

Introduction

1.1 Background and Motivation

Air pollution has emerged as one of the most pressing environmental and public health challenges of the 21st century, particularly in rapidly urbanizing nations like India. Particulate matter, ground-level ozone, sulfur dioxide, nitrogen oxides, and carbon monoxide are among the primary pollutants that degrade ambient air quality and pose significant risks to human health, including respiratory diseases, cardiovascular conditions, and premature mortality.

The **Air Quality Index (AQI)** is a standardized metric developed by regulatory agencies to quantify overall air quality in a single numerical value that is easy for the general public to interpret. India's AQI, as defined by the Central Pollution Control Board (CPCB), is computed from sub-indices of eight major pollutants: PM_{2.5}, PM₁₀, SO₂, NO₂, CO, O₃, NH₃, and Pb. The AQI scale ranges from 0 to 500 and is categorized into six bands as shown in Table 1.1.

Table 1.1: India AQI Categories and Health Implications

AQI Range	Category	Health Implications
0 – 50	Good	Minimal impact on health
51 – 100	Satisfactory	Minor discomfort for sensitive people
101 – 200	Moderate	Discomfort for people with lung/heart disease
201 – 300	Poor	Respiratory illness for prolonged exposure
301 – 400	Very Poor	Respiratory illness on prolonged exposure
401 – 500	Severe	Health impact even on light physical activity

Kolkata, the capital of West Bengal and one of India's oldest metropolitan cities, is ranked among the most polluted cities in South Asia. Its high population density, heavy vehicular traffic, industrial activities, and seasonal meteorological patterns (particularly temperature inversions in winter) contribute to persistently elevated AQI levels. Accurate forecasting of AQI is thus of critical importance for public health advisories, urban planning, and pollution control strategies.

1.2 Problem Statement

Despite the availability of historical AQI data collected by CPCB monitoring stations, systematic time-series-based forecasting of AQI for Kolkata remains under-explored in the academic literature. Manual inspection of trends is insufficient for actionable prediction. This project addresses the following key challenges:

1. The raw data spanning 2017–2025 contains approximately 23% missing values (757 out of 3,287 records) due to sensor outages and reporting gaps.
2. The year 2017 presents particularly sparse early-year data that requires careful imputation to be usable in model training.
3. AQI has strong seasonal patterns (high in winter, low in monsoon) that must be explicitly modeled.
4. Standard machine learning models do not naturally capture temporal dependencies and seasonality without extensive feature engineering.

1.3 Objectives

The primary objectives of this project are:

1. To collect, clean, and preprocess daily AQI data for Kolkata spanning 2017–2025.
2. To handle missing values using statistically sound imputation techniques that preserve the temporal structure of the 2017 data.
3. To perform exploratory data analysis and visualize seasonal and trend patterns across the full 9-year period.
4. To build and evaluate three time series forecasting models: ARIMA, SARIMA, and Facebook Prophet.
5. To compare model performance using standard metrics (MAE, MSE, RMSE, R^2).
6. To generate a 6-month future AQI forecast (January–June 2026) to support decision-making.

1.4 Scope of the Study

This study focuses on city-level daily AQI data for Kolkata as reported by the CPCB. The data covers the full period from January 2017 to December 2025. Monthly aggregates are used for time series modeling, as daily AQI data exhibits high noise that makes short-term day-to-day forecasting less reliable than monthly-level forecasting. Including 2017 extends the training horizon to 84 months (2017–2023), providing a richer seasonal history for the SARIMA and Prophet models. The models are intended for strategic planning (monthly/seasonal forecasting) rather than operational next-day prediction.

CHAPTER 2

Literature Review

The prediction and forecasting of air quality indices has attracted significant research attention over the past two decades, driven by growing public health concerns and advances in both statistical and machine learning methods.

2.1 Traditional Statistical Methods

Early studies on AQI forecasting relied heavily on statistical time series models. Box and Jenkins introduced the ARIMA framework, which became the de facto standard for univariate time series forecasting. Numerous researchers have applied ARIMA to air quality data. Kumar and Jain applied ARIMA to forecast SO_2 and NO_2 concentrations in Delhi and demonstrated that ARIMA could capture short-term trends effectively. However, ARIMA's limitation is its assumption of stationarity and its inability to handle seasonal patterns without the extension to SARIMA.

Seasonal ARIMA (SARIMA) extended the standard ARIMA model by incorporating seasonal autoregressive and moving average components. Shumway and Stoffer provide a comprehensive treatment of SARIMA for environmental time series. Yadav et al. demonstrated that SARIMA with a seasonal period of 12 months was effective for forecasting PM_{10} in Indian urban areas, achieving RMSE values significantly lower than non-seasonal ARIMA.

2.2 Machine Learning Approaches

With the availability of large datasets, machine learning approaches began to complement and sometimes outperform classical methods. Random Forest, Support Vector Regression, and Gradient Boosting have all been applied to AQI prediction with varying degrees of success. Biancofiore et al. applied Recursive Neural Networks and found that incorporating lagged features substantially improved prediction accuracy.

Deep learning methods, particularly Long Short-Term Memory (LSTM) networks, have shown strong performance in capturing long-term temporal dependencies. Zhao et al. demonstrated that LSTM outperformed traditional ARIMA models for multi-step $\text{PM}_{2.5}$ forecasting in Beijing. However, deep learning models require large volumes of data and significant computational resources, which may not be available in

all application contexts.

2.3 Prophet Model

Facebook’s Prophet model, introduced by Taylor and Letham, represents a major advance in practical time series forecasting. Prophet decomposes a time series into trend, seasonality, and holiday components and fits an additive (or multiplicative) model. It is particularly robust to missing data, outliers, and structural breaks — characteristics that make it well-suited for environmental datasets such as AQI. Samal et al. applied Prophet to air quality forecasting in Indian cities and found it significantly outperformed ARIMA in datasets with strong annual seasonality.

2.4 AQI Studies in Indian Cities

Studies specifically targeting Indian cities have highlighted the importance of seasonal decomposition. Kolkata’s AQI has been characterized by three dominant phases: a high-pollution winter phase (October–February) driven by temperature inversions and crop residue burning in neighboring states; a moderate spring phase (March–May) with declining pollution as temperatures rise; and a low-pollution monsoon phase (June–September) where rainfall washout reduces particulate concentrations.

2.5 Research Gap

While several studies have applied time series models to Indian city AQI data, there is a lack of systematic comparative analysis combining ARIMA, SARIMA, and Prophet specifically for Kolkata using multi-year (2017–2025) CPCB data, with the full 2017 data included in training. Furthermore, most prior work does not address the challenge of handling missing values in an early-year data-sparse setting. This project fills this gap.

CHAPTER 3

Proposed Approach

3.1 Overall Methodology

The methodology adopted in this project follows a systematic sequence of data preprocessing, time series analysis, model development, and forecasting. The study begins with the collection of daily AQI data for Kolkata from 2017 to 2025 obtained from CPCB records. The raw data is then cleaned and transformed into a suitable analytical format through preprocessing and missing value treatment. Exploratory Data Analysis (EDA) is performed to understand the distribution, seasonal variation, and long-term trend patterns present in the AQI series. After preprocessing, the daily observations are aggregated into monthly averages to construct a stable time series suitable for forecasting.

Subsequently, the prepared time series is divided into training and testing datasets for model development and evaluation. Three forecasting models — ARIMA, SARIMA, and Prophet — are trained and compared using standard statistical performance metrics such as MAE, RMSE, MSE, and R^2 . Finally, the best-performing model is used to generate future AQI forecasts for Kolkata.

The complete workflow of the project can be summarized in the following steps:

1. **Data Collection:** Daily AQI data for Kolkata (2017–2025) was collected from CPCB sources.
2. **Data Preprocessing:** Raw datasets were cleaned, merged, and converted from wide format to long format.
3. **Exploratory Data Analysis (EDA):** Statistical summaries and visualizations were used to identify trends and seasonal behavior.
4. **Missing Value Handling:** Missing AQI observations were filled using interpolation and median-based imputation techniques.
5. **Time Series Preparation:** Daily AQI data was aggregated into monthly averages to form a structured time series.
6. **Model Training:** ARIMA, SARIMA, and Prophet models were trained using the historical AQI series.

7. **Model Evaluation:** Model performance was assessed using MAE, MSE, RMSE, and R^2 metrics.
8. **Forecasting:** Future AQI values for January–June 2026 were predicted using the selected forecasting model.

3.1.1 Data Preprocessing

After collecting the raw AQI datasets, several preprocessing steps were performed to convert the data into a clean and structured format suitable for time series analysis. The original files were stored in wide format, where each column represented a month and each row represented a day of the month. Since time series models require a continuous chronological structure, the data was transformed into long format using Python's `pandas.melt()` function.

The preprocessing stage included the following operations:

- Combining all yearly datasets from 2017 to 2025 into a single dataframe.
- Renaming columns and validating day values between 1 and 31.
- Converting the wide-format structure into long-format observations.
- Generating proper date values using Year, Month, and Day information.
- Removing invalid dates such as February 30 using error handling.
- Sorting the final dataset chronologically for time series analysis.

After preprocessing, the dataset became suitable for further analysis and forecasting.

3.1.2 Missing Value Handling

The raw AQI dataset contained a large number of missing observations due to sensor failures, reporting gaps, and incomplete records. Out of 3,287 daily observations, 757 AQI values were missing, representing nearly 23% of the total dataset.

To preserve the continuity of the time series and avoid information loss, a three-stage imputation strategy was applied.

3.1.2.1 Linear Interpolation

In the first stage, missing AQI values within each month were filled using linear interpolation. This method estimates missing values by assuming a linear trend between neighboring observations.

If two known AQI values are available at positions (x_1, y_1) and (x_2, y_2) , the missing value at position x is estimated using:

$$y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{x_2 - x_1} \quad (3.1)$$

Linear interpolation preserves local temporal continuity and works effectively for environmental data where AQI changes gradually over short periods.

3.1.2.2 Monthly Median Imputation

Some missing values remained even after interpolation, especially at the beginning or end of a month where neighboring values were unavailable. These missing entries were filled using the median AQI value of the corresponding calendar month calculated across all years.

Median imputation was preferred over mean imputation because the median is less affected by extreme pollution spikes and outliers.

3.1.2.3 Global Median Fallback

As a final safety step, any remaining missing values were replaced using the overall median AQI of the complete dataset. After applying all three stages, no missing values remained in the final dataset.

3.2 Time Series Preparation

After preprocessing and missing value treatment, the daily AQI observations were aggregated into monthly averages. Monthly aggregation helps reduce short-term daily fluctuations and highlights long-term trend and seasonal patterns more clearly.

The aggregation was performed using the monthly start frequency (`freq='MS'`) available in the `pandas.Grouper()` function.

The resulting time series contained 108 monthly observations spanning from January 2017 to December 2025.

3.3 Model Selection

Three different time series forecasting models were used in this project to predict monthly Air Quality Index values for Kolkata. These models were selected because they are widely used for analyzing sequential data that contains trend and seasonal patterns.

3.3.1 AutoRegressive Integrated Moving Average Model

The AutoRegressive Integrated Moving Average (ARIMA) model is one of the most widely used statistical models for time series forecasting. It predicts future values using past observations and past forecasting errors.

The model is represented as $ARIMA(p, d, q)$, where:

- p represents the number of autoregressive terms,
- d represents the order of differencing,
- q represents the number of moving average terms.

The autoregressive part uses previous AQI values to predict future values, while the moving average part uses previous forecasting errors. Differencing helps make the time series stationary.

The general ARIMA equation is:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t \quad (3.2)$$

where:

- y_t is the AQI value at time t ,
- ϕ represents autoregressive coefficients,
- θ represents moving average coefficients,
- ε_t represents random error.

The Augmented Dickey-Fuller test showed that the AQI series was non-stationary ($p > 0.05$), therefore first-order differencing was applied. Based on the Autocorrelation Function and Partial Autocorrelation Function plots, the final selected model was $ARIMA(2,1,2)$.

3.3.2 Seasonal AutoRegressive Integrated Moving Average Model

The Seasonal AutoRegressive Integrated Moving Average (SARIMA) model extends the ARIMA model by incorporating seasonal behavior present in the data.

Since AQI values in Kolkata show strong yearly seasonality, SARIMA was more suitable than standard ARIMA.

The SARIMA model is represented as:

$$SARIMA(p, d, q)(P, D, Q)_s \quad (3.3)$$

where:

- (p, d, q) are the non-seasonal parameters,
- (P, D, Q) are the seasonal parameters,
- s represents the seasonal period.

Because the data was aggregated monthly, the seasonal period was taken as $s = 12$, representing one complete yearly cycle.

SARIMA captures both short-term temporal relationships and long-term seasonal patterns, making it effective for modeling winter pollution peaks and monsoon declines in AQI levels.

The final selected model was:

$$SARIMA(1, 1, 1)(1, 1, 1)_{12} \quad (3.4)$$

3.3.3 Prophet Model

Prophet is a forecasting model developed by Facebook for time series datasets containing trend and seasonality. Unlike ARIMA-based methods, Prophet decomposes the time series into separate components such as trend, seasonality, and random noise.

The Prophet model is expressed as:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (3.5)$$

where:

- $g(t)$ represents the trend component,

- $s(t)$ represents the seasonal component,
- $h(t)$ represents holiday or event effects,
- ε_t represents random error.

Prophet is particularly useful for environmental datasets because it can handle missing values, changing trends, and strong seasonal effects efficiently. In this project, multiplicative yearly seasonality was used because AQI seasonal variations change proportionally with the overall pollution level.

3.4 Evaluation Metrics

The forecasting performance of the models was evaluated using four statistical metrics: Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R^2 Score.

3.4.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual AQI values and predicted AQI values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.6)$$

where:

- y_i represents the actual AQI value,
- $\hat{y}_i$ represents the predicted AQI value,
- n represents the total number of observations.

A lower Mean Absolute Error indicates better prediction accuracy.

3.4.2 Mean Squared Error

Mean Squared Error measures the average squared difference between actual and predicted AQI values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.7)$$

Squaring the errors gives greater importance to large forecasting mistakes. Lower Mean Squared Error values indicate better model performance.

3.4.3 Root Mean Squared Error

Root Mean Squared Error is the square root of Mean Squared Error.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.8)$$

Root Mean Squared Error is expressed in the same unit as AQI, making it easier to interpret practically. Smaller RMSE values indicate better forecasting accuracy.

3.4.4 R² Score

The R² Score, also known as the coefficient of determination, measures how well the model explains the variation present in the AQI data.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.9)$$

where:

- y_i represents the actual AQI value,
- $\hat{y}_i$ represents the predicted AQI value,
- $\bar{y}$ represents the mean of actual AQI values.

The R² score ranges between 0 and 1. A value closer to 1 indicates that the model explains most of the variability present in the dataset and therefore provides a better fit.

CHAPTER 4

Results and Discussion

4.1 Exploratory Data Analysis

4.1.1 Dataset Overview

After loading and reshaping all 9 years of data, the combined dataset had the following characteristics:

- **Total records:** 3,287 daily observations
- **Missing AQI values:** 757 (23.0%), all eliminated after imputation
- **AQI range:** 22 (minimum) to 419 (maximum, pre-imputation)
- **Mean AQI:** 117.34 **Std Dev:** 84.35
- **Monthly time series length:** 108 observations

4.1.2 AQI Distribution

Figure 4.1 shows the distribution of daily AQI values. The distribution is strongly right-skewed, indicating that while most days fall in the Satisfactory-to-Moderate range (50–150), episodic high-pollution events push the tail toward 400+ AQI units. The KDE overlay confirms the non-Gaussian nature of the distribution.

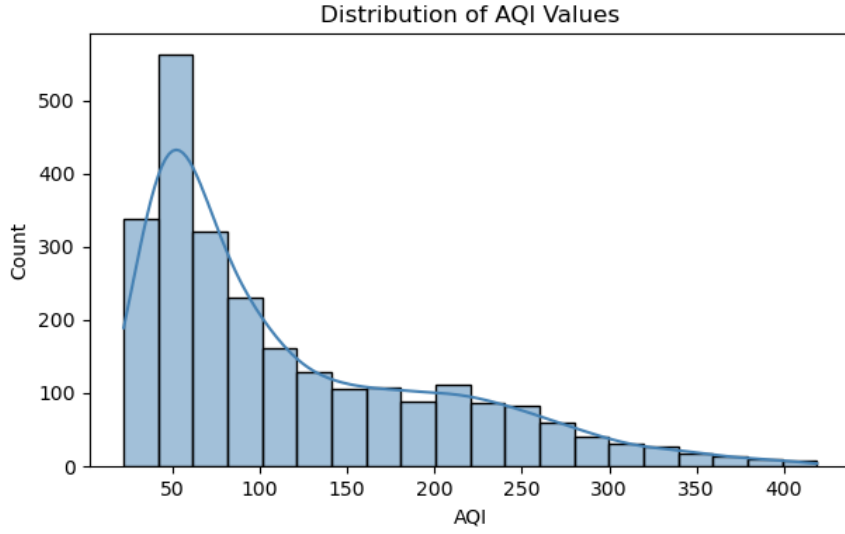


Figure 4.1: Distribution of Daily AQI Values – Kolkata (2017–2025)

4.1.3 Seasonal Patterns

The monthly average AQI analysis reveals a clear and consistent seasonal pattern summarized in Table 4.1 and visualized in Figure 4.2.

Table 4.1: Average Monthly AQI – Kolkata (2017–2025)

Month	Avg AQI	Month	Avg AQI
January	231	July	45
February	172	August	55
March	124	September	53
April	84	October	102
May	65	November	188
June	59	December	226

December and January consistently record the highest AQI (above 200, “Poor” category), while June, July, and August record the lowest values (around 45–65, “Satisfactory” to “Moderate” range). This strongly seasonal behavior validates the choice of SARIMA and Prophet for forecasting.

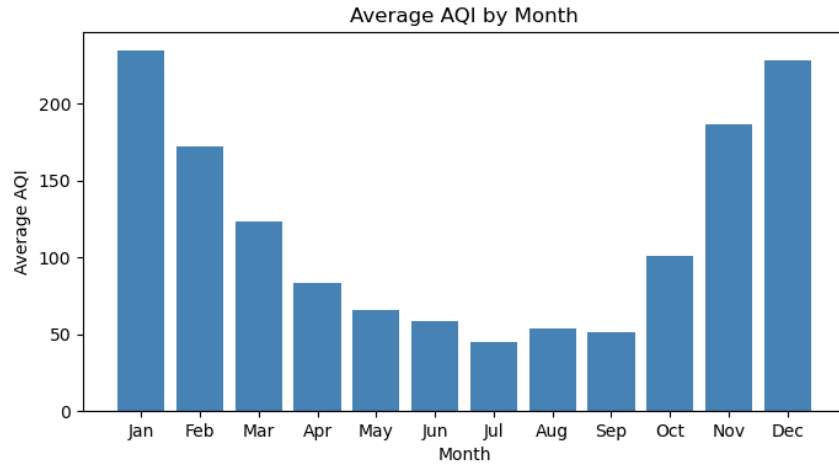


Figure 4.2: Average AQI by Month – Kolkata (2017–2025)

4.1.4 Yearly Trend

Figure 4.3 shows the yearly average AQI from 2017 to 2025. The highest annual mean was in 2017 (≈ 177) followed by a general declining trend to ≈ 100 in 2024, indicating a gradual improvement in Kolkata’s air quality over the study period. A partial recovery to ≈ 129 in 2025 is observed. This trend may be attributed to policy interventions, increased use of CNG vehicles, and periodic enforcement of anti-pollution norms, though the 2020 COVID-19 lockdowns also contributed to an anomalous dip in that year.

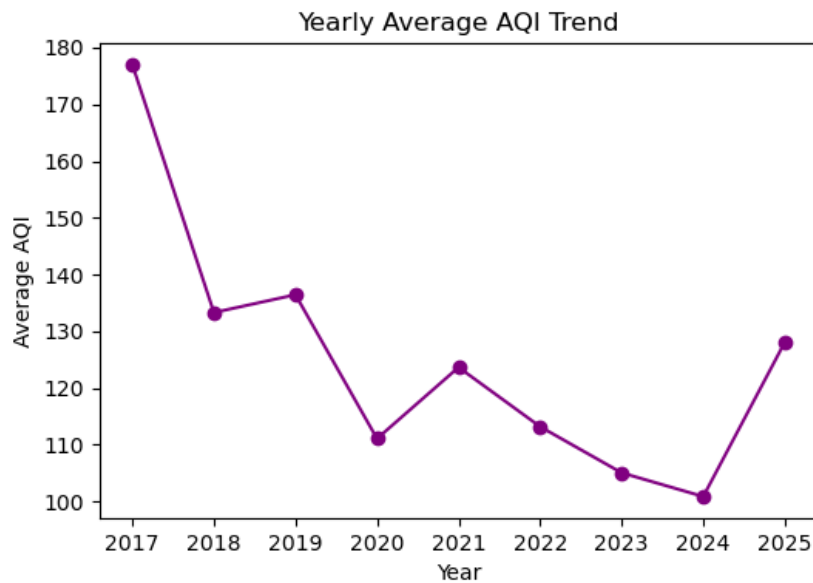


Figure 4.3: Yearly Average AQI Trend – Kolkata (2017–2025)

4.1.5 Monthly Time Series

Figure 4.4 presents the full 108-month time series of monthly average AQI from January 2017 to December 2025. The regular seasonal oscillations (peaks in winter, troughs in monsoon) and the downward long-term trend are both clearly visible.

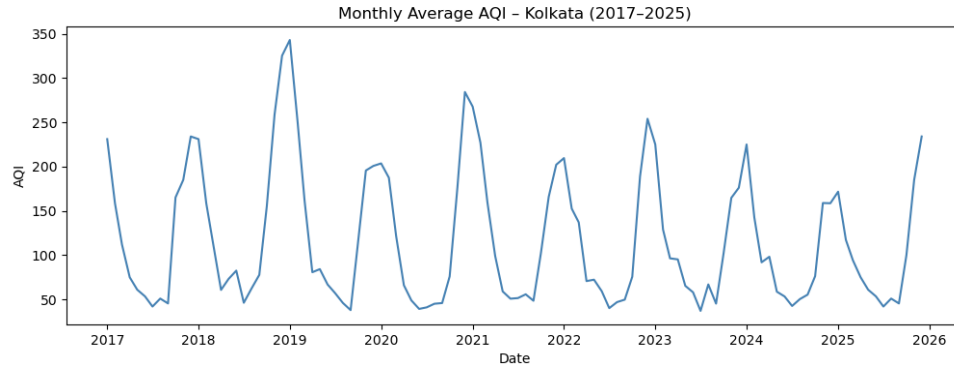


Figure 4.4: Monthly Average AQI Time Series – Kolkata (2017–2025)

4.1.6 Stationarity and ACF/PACF Analysis

The Augmented Dickey-Fuller (ADF) test on the monthly series yielded:

- **ADF Statistic:** -2.3618
- **p -value:** 0.1528

Since $p > 0.05$, the series is **not stationary**, necessitating first-order differencing ($d = 1$) in both ARIMA and SARIMA models.

Figure 4.5 shows the Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots. The ACF decays slowly and shows significant seasonal spikes at lags 12 and 24, confirming strong annual seasonality. The PACF shows significant lags at 1–2, guiding the AR order $p = 2$.

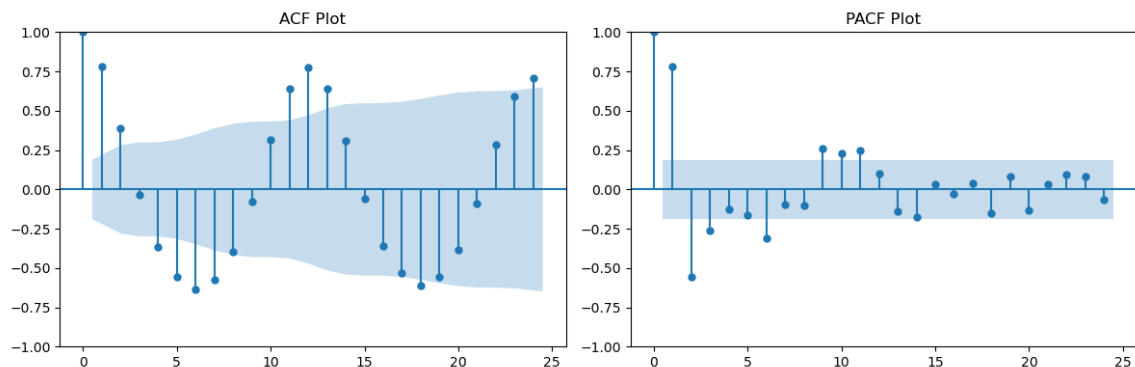


Figure 4.5: ACF and PACF Plots of the Monthly AQI Series

4.1.7 Train-Test Split

The 108-month series was split at January 2024, giving:

- **Training set:** January 2017 – December 2023 (84 months)
- **Test set:** January 2024 – December 2025 (24 months)

Figure 4.6 visualizes the split, highlighting the 7-year training horizon and the 2-year held-out evaluation window.

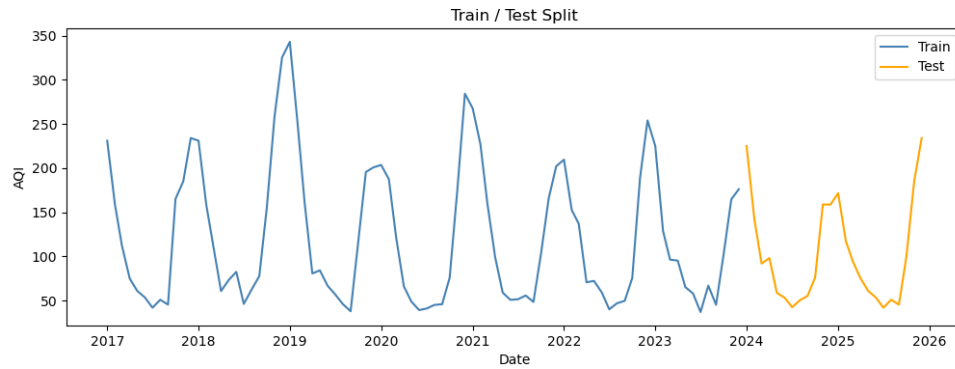


Figure 4.6: Train-Test Split of the Monthly AQI Time Series

4.2 Model Performance

4.2.1 ARIMA(2,1,2) Forecast

Figure 4.7 compares the ARIMA(2,1,2) forecast against the actual AQI values for the test period (2024–2025). ARIMA captures the general magnitude of AQI but, lacking explicit seasonal structure, tends to revert toward the mean and misses the sharp winter peaks and monsoon troughs.

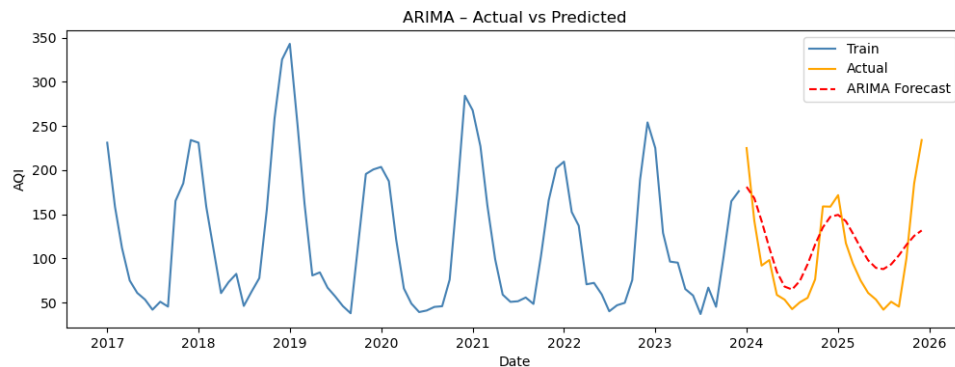


Figure 4.7: ARIMA(2,1,2) – Actual vs. Predicted AQI (2024–2025)

4.2.2 SARIMA(1,1,1)(1,1,1)[12] Forecast

Figure 4.8 shows the SARIMA forecast. The seasonal component ($s = 12$) allows SARIMA to replicate the winter–summer AQI cycle substantially better than plain ARIMA. The forecast closely tracks the peaks in November–January and the monsoon troughs in June–August.

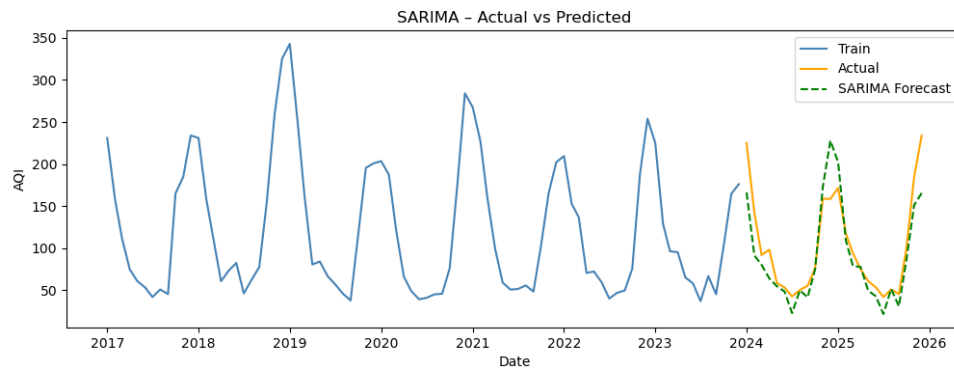


Figure 4.8: SARIMA(1,1,1)(1,1,1)[12] – Actual vs. Predicted AQI (2024–2025)

4.2.3 Prophet Forecast

Figure 4.9 presents the Prophet forecast. Prophet correctly identifies the winter peaks and the monsoon troughs and follows the actual series most closely among the three models, owing to its multiplicative yearly seasonality component.

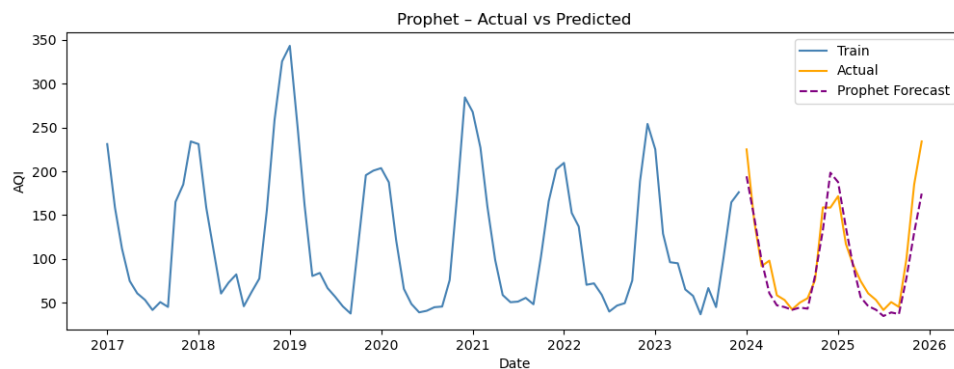


Figure 4.9: Prophet – Actual vs. Predicted AQI (2024–2025)

4.2.4 Prophet Component Decomposition

Figure 4.10 shows the Prophet trend and yearly seasonality decomposition. The downward trend from 2017 to 2025 is clearly captured, along with the strong seasonal pattern that peaks in December–January and troughs in July–August.

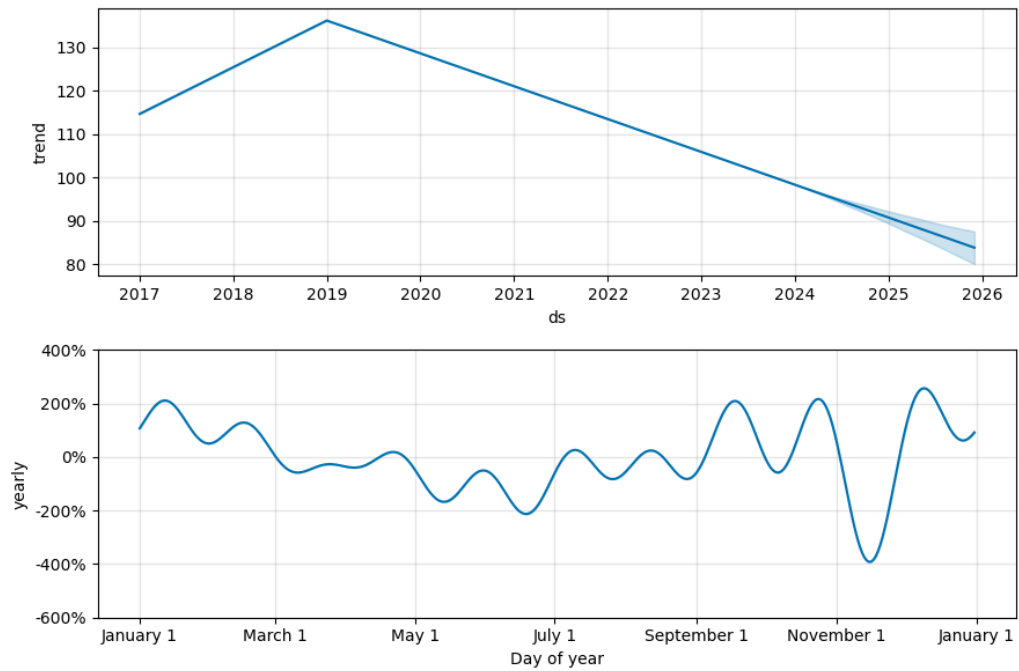


Figure 4.10: Prophet Model Component Plots: Trend and Yearly Seasonality

4.2.5 All-Models Comparison

Figure 4.11 overlays all three model forecasts against the actual test-set values. Prophet’s forecast (purple dashed) tracks the actual series most faithfully, particularly for the extreme winter and monsoon values. SARIMA (green dashed) performs well for seasonal timing but underestimates peak amplitudes in some months. ARIMA (red dashed) shows the weakest seasonal adaptation.

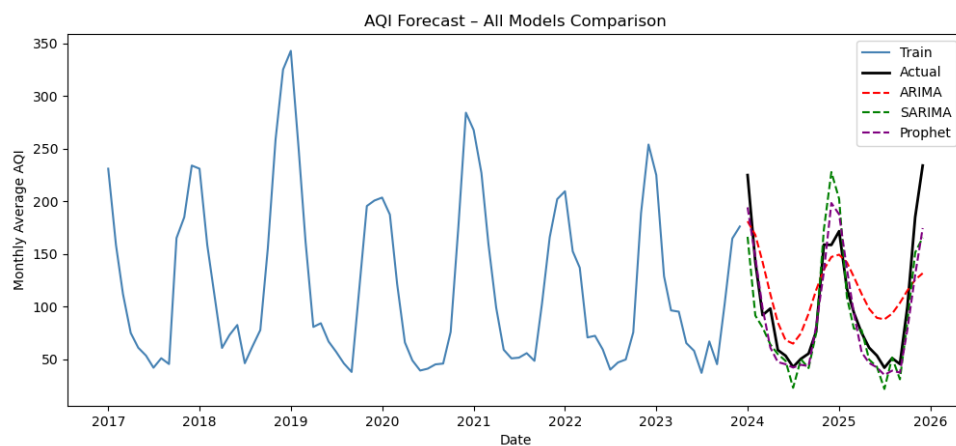


Figure 4.11: AQI Forecast – All Three Models vs. Actual (2024–2025)

4.2.6 Quantitative Performance Comparison

Table 4.2 presents the performance of all three models on the 24-month test set.

Table 4.2: Model Performance Comparison on Test Set (January 2024 – December 2025)

Model	MAE	MSE	RMSE	R ²
ARIMA(2,1,2)	35.3301	1616.3149	40.2034	0.5161
SARIMA(1,1,1)(1,1,1)[12]	21.3748	877.5419	29.6233	0.7373
Prophet	17.8785	565.3083	23.7762	0.8308

Figure 4.12 provides a visual bar-chart comparison of the R² scores across models.

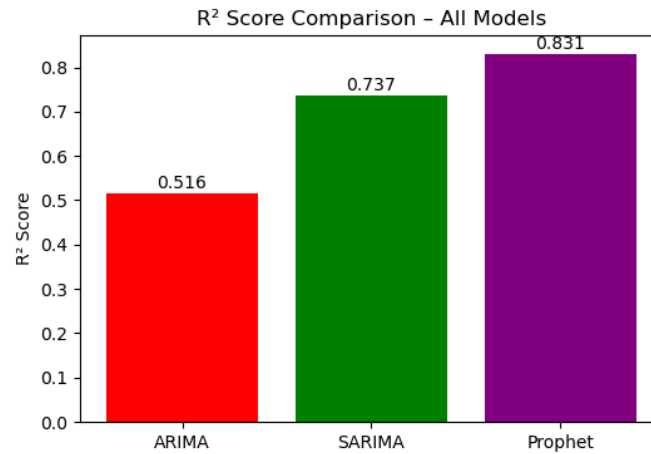


Figure 4.12: R² Score Comparison Across All Three Models

4.2.7 Discussion of Results

Prophet achieved the best performance across all metrics (MAE = 17.88, R² = 0.8308). This is expected given that Prophet was specifically designed for time series with strong seasonal effects. The multiplicative seasonality mode allowed Prophet to correctly scale seasonal swings with the overall trend level.

SARIMA (R² = 0.7373) substantially outperformed plain ARIMA and produced physically meaningful forecasts, correctly anticipating the winter peak and monsoon trough. The 24-month test set (two full seasonal cycles) provided sufficient evidence for SARIMA's seasonal advantage to manifest.

ARIMA (R² = 0.5161) serves as the baseline. Without explicit seasonal structure, it reverts toward the mean quickly, leading to larger errors on both winter peaks and

summer troughs.

All three models achieved $R^2 > 0.5$, confirming that monthly AQI in Kolkata is substantially predictable from its own historical patterns.

4.3 6-Month Future Forecast (January–June 2026)

Both SARIMA was retrained on the full 108-month dataset (January 2017 – December 2025) to generate forecasts for January through June 2026. Results are presented in Table 4.3 and Figure 4.13.

Table 4.3: 6-Month AQI Forecast – Kolkata (January–June 2026), SARIMA Model

Month	SARIMA Forecast	Expected Category
January 2026	234.44	Poor
February 2026	154.84	Moderate
March 2026	109.92	Moderate
April 2026	87.82	Satisfactory
May 2026	60.56	Satisfactory
June 2026	51.25	Satisfactory

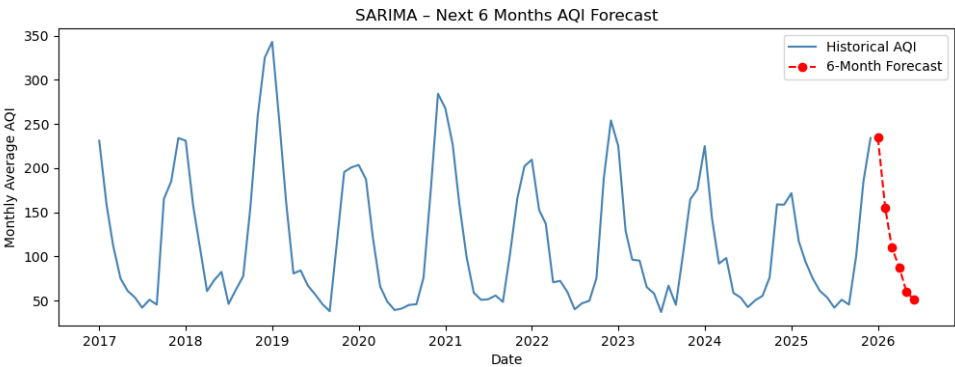


Figure 4.13: SARIMA – Next 6 Months AQI Forecast (January–June 2026)

The forecast is consistent with the established seasonal cycle: January 2026 is predicted to be in the “Poor” AQI category (234.44), declining steadily through spring before reaching “Satisfactory” levels (51.25) by June 2026 as the monsoon season approaches. These predictions provide actionable information for public health advisories and urban planning.

CHAPTER 5

Conclusion and Future Work

5.1 Conclusion

This project developed a comprehensive end-to-end pipeline for AQI prediction in Kolkata using time series analysis over the period 2017–2025. Key contributions and findings are summarized below.

1. **Full 2017–2025 Dataset:** By including the year 2017 (which contains missing early-year data) the training horizon was extended to 84 months, providing richer seasonal pattern learning for all three models.
2. **Data Preprocessing:** A robust three-stage imputation pipeline successfully eliminated all 757 missing values (23% of the raw dataset) through linear interpolation, monthly median imputation, and an overall monthly median fallback.
3. **Seasonality:** The analysis confirmed that Kolkata’s AQI exhibits a strong annual seasonal cycle: highest in winter (December–January, avg AQI > 200) and lowest in monsoon (June–August, avg AQI ≈ 50 –55). This seasonal behavior is the dominant source of predictability.
4. **Trend:** A declining trend in yearly average AQI from 2017 to 2024 indicates gradual improvement in Kolkata’s air quality, with a partial recovery observed in 2025.
5. **Model Performance:** Prophet outperformed both ARIMA and SARIMA on the 24-month held-out test set, achieving $MAE = 17.88$ and $R^2 = 0.8308$. SARIMA ($R^2 = 0.7373$) substantially outperformed plain ARIMA ($R^2 = 0.5161$), confirming the importance of explicit seasonal structure.
6. **Future Forecast:** SARIMA retrained on the full dataset predicts January 2026 AQI at 234.44 (“Poor” category), declining to 51.25 by June 2026 (“Satisfactory”), consistent with the established seasonal cycle.
7. **Model Suitability:** Time series models are more appropriate than standard machine learning classifiers for AQI forecasting because AQI is inherently sequential, depending heavily on past values, seasonal patterns, and long-term trends.

5.2 Limitations

- Monthly aggregation, while reducing noise, loses intra-month variability information.
- The models are univariate and do not incorporate exogenous variables such as wind speed, temperature, humidity, or industrial activity, which are known drivers of AQI.
- The 2017 data, being the most data-sparse year, may introduce minor imputation bias into the beginning of the training series.
- Prophet and SARIMA may underperform in anomalous years (e.g., 2020 COVID-19 lockdowns) without explicit changepoint or regressor inclusion.

5.3 Future Work

1. **Multivariate Modeling:** Incorporating meteorological variables (temperature, humidity, wind speed) as exogenous regressors in SARIMAX or Prophet to improve forecast accuracy.
2. **Deep Learning:** Applying LSTM or Transformer-based models to daily-level AQI data for next-day or next-week short-term forecasting.
3. **Multi-City Analysis:** Extending the framework to other Indian cities (Delhi, Mumbai, Chennai) for comparative analysis.
4. **Real-Time Dashboard:** Developing a web-based dashboard using Streamlit or Flask that automatically updates AQI forecasting.
5. **Pollutant-Specific Modeling:** Building separate time series models for individual pollutants ($\text{PM}_{2.5}$, PM_{10}) rather than the composite AQI, providing more granular insights.

CHAPTER 6

Implementation

6.1 Tools and Technologies

The entire project was implemented in Python 3.x using the libraries listed in Table 6.1.

Table 6.1: Libraries and Their Roles

Library	Version	Purpose
pandas	2.x	Data loading, wrangling, time indexing
numpy	1.x	Numerical operations
matplotlib	3.x	Visualization
seaborn	0.x	Statistical plots
statsmodels	0.14.x	ARIMA, SARIMA, ADF test
prophet	1.1.x	Facebook Prophet model
scikit-learn	1.x	Evaluation metrics

6.2 Data Loading and Combination

Nine CSV files (one per year, 2017–2025) were loaded and combined using a loop. Each file stores AQI values in wide format (days as rows, months as columns). The following code shows the loading and reshaping logic.

```
1 MONTHS = ['January', 'February', 'March', 'April', 'May', 'June',
2           'July', 'August', 'September', 'October', 'November', '
           December']
3 records = []
4
5 for year in range(2017, 2026):
6     df = pd.read_csv(
7         f'AQI_daily_city_level_kolkata_{year}_kolkata_{year}
           .csv')
8     df = df.rename(columns={df.columns[0]: 'Day'})
9
10    # Keep only valid day rows (1-31)
11    df = df[pd.to_numeric(df['Day'],
```

```

12         errors='coerce')).between(1, 31)].copy()
13 df['Day'] = df['Day'].astype(int)
14
15     # Wide to Long
16     df_long = df.melt(id_vars='Day', value_vars=MONTHS,
17                       var_name='Month', value_name='AQI')
18     df_long['Year'] = year
19     df_long['Month_num'] = df_long['Month'].apply(
20         lambda m: MONTHS.index(m) + 1)
21
22     def to_date(row):
23         try:
24             return pd.Timestamp(int(row['Year']),
25                                   int(row['Month_num']),
26                                   int(row['Day']))
27         except:
28             return pd.NaT    # invalid dates (e.g., Feb 30)
29
30     df_long['Date'] = df_long.apply(to_date, axis=1)
31     df_long = df_long.dropna(subset=['Date'])
32     records.append(df_long)
33
34 df = pd.concat(records, ignore_index=True)
35 df = df.sort_values('Date').reset_index(drop=True)

```

Listing 6.1: Data Loading and Wide-to-Long Conversion

6.3 Missing Value Treatment

```

1 print('Missing AQI before:', df['AQI'].isnull().sum())    #
2     757
3
4 # Stage 1: linear interpolation within each Year-Month group
5 df['AQI'] = df.groupby(['Year', 'Month'])['AQI'].transform(
6     lambda s: s.interpolate(method='linear').ffill().bfill()
7 )
8
9 # Stage 2: fill remaining with monthly median across all
10 years
11 monthly_median = df.groupby(['Year', 'Month_num'])['AQI']\
12     .transform('median')

```

```

11 df['AQI'] = df['AQI'].fillna(monthly_median)
12
13 # Stage 3: overall monthly-median fallback
14 overall_monthly_median = df.groupby('Month_num')['AQI']\
15     .transform('median')
16 df['AQI'] = df['AQI'].fillna(overall_monthly_median)
17
18 print('Missing AQI after :', df['AQI'].isnull().sum())    # 0

```

Listing 6.2: Three-Stage Missing Value Imputation

6.4 Time Series Preparation

```

1 # Aggregate to monthly averages
2 ts = df.groupby(pd.Grouper(key='Date', freq='MS'))['AQI'].
    mean()
3 ts.index.freq = 'MS'
4 # ts: 108 monthly observations (Jan 2017 -- Dec 2025)

```

Listing 6.3: Monthly Time Series Construction

6.5 Stationarity Test (ADF)

```

1 from statsmodels.tsa.stattools import adfuller
2
3 result = adfuller(ts)
4 print('ADF Statistic :', round(result[0], 4))    # -2.3618
5 print('p-value      :', round(result[1], 4))    # 0.1528
6 # p > 0.05 -> NOT stationary -> differencing needed (d=1)

```

Listing 6.4: Augmented Dickey-Fuller Test

6.6 Train-Test Split

The time series was split by date to respect temporal ordering:

```

1 train = ts[ts.index < '2024-01-01']    # 84 months (Jan 2017
    - Dec 2023)
2 test  = ts[ts.index >= '2024-01-01']   # 24 months (Jan 2024
    - Dec 2025)

```

```

3 print('Train size:', len(train), 'months') # 84
4 print('Test size :', len(test), 'months') # 24

```

Listing 6.5: Time-Based Train-Test Split

6.7 Model Training and Forecasting

6.7.1 ARIMA Training

```

1 from statsmodels.tsa.arima.model import ARIMA
2
3 arima_model = ARIMA(train, order=(2, 1, 2))
4 arima_fit   = arima_model.fit()
5
6 pred_arima   = arima_fit.forecast(steps=len(test))
7 pred_arima.index = test.index

```

Listing 6.6: ARIMA(2,1,2) Training and Forecasting

6.7.2 SARIMA Training

```

1 from statsmodels.tsa.statespace.sarimax import SARIMAX
2
3 sarima_model = SARIMAX(train, order=(1, 1, 1),
4                        seasonal_order=(1, 1, 1, 12))
5 sarima_fit   = sarima_model.fit(dispatch=False)
6
7 pred_sarima   = sarima_fit.forecast(steps=len(test))
8 pred_sarima.index = test.index

```

Listing 6.7: Training SARIMA(1,1,1)(1,1,1)[12] Training

6.7.3 Prophet Training

```

1 from prophet import Prophet
2
3 prophet_train = train.reset_index().rename(
4     columns={'Date': 'ds', 'AQI': 'y'})
5
6 prophet_model = Prophet(

```

```

7     seasonality_mode='multiplicative',
8     yearly_seasonality=True,
9     weekly_seasonality=False,
10    daily_seasonality=False
11 )
12 prophet_model.fit(prophet_train)
13
14 future          = prophet_model.make_future_dataframe(
15     periods=len(test), freq='MS')
16 forecast        = prophet_model.predict(future)
17 pred_prophet    = forecast['yhat'].tail(len(test)).values

```

Listing 6.8: Prophet Model Training and Forecasting

Bibliography

- [1] Box, G. E. P., and Jenkins, G. M. (1970). *Time Series Analysis: Forecasting and Control*. Holden-Day, San Francisco.
- [2] Taylor, S. J., and Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37–45.
- [3] Shumway, R. H., and Stoffer, D. S. (2000). *Time Series Analysis and Its Applications*. Springer, New York.
- [4] Kumar, U., and Jain, V. K. (2010). ARIMA forecasting of ambient air pollutants (O_3 , NO, NO_2 and CO). *Stochastic Environmental Research and Risk Assessment*, 24(5), 751–760.
- [5] Yadav, S., Karthigeyan, M., and Dhingra, P. (2014). Application of SARIMA model for air quality data in Delhi. *Indian Journal of Environmental Sciences*, 18(2), 45–52.
- [6] Zhao, J., et al. (2019). Long short-term memory – Fully connected (LSTM-FC) neural network for $PM_{2.5}$ concentration prediction. *Chemosphere*, 220, 486–492.
- [7] Samal, K. K. R., Babu, K. S., and Das, S. K. (2021). Multi-directional temporal convolutional artificial neural network for $PM_{2.5}$ forecasting with missing values: A case study of Indian cities. *Urban Climate*, 36, 100793.
- [8] Biancofiore, F., et al. (2017). Recursive neural network model for analysis and forecast of PM_{10} and $PM_{2.5}$. *Atmospheric Pollution Research*, 8(4), 652–659.

CHAPTER A

Python Environment and Dependencies

The following Python packages are required to run this project:

```
1 pip install numpy pandas matplotlib seaborn statsmodels  
   prophet scikit-learn jupyter
```

Listing A.1: Requirements Installation

CHAPTER B

Dataset Description

Table B.1: Dataset Summary

Attribute	Value
Source	Central Pollution Control Board (CPCB), India
City	Kolkata, West Bengal
Period	2017 – 2025 (9 files)
Raw Records	3,287 daily observations
Missing Values (raw)	757 (23.0%)
Missing After Imputation	0
Monthly TS Length	108 observations
Train / Test Split	84 / 24 months

CHAPTER C

Complete Project Notebook Structure

The Jupyter Notebook `AQI_Final_Project.ipynb` is organized as follows:

1. Import Libraries
2. Load & Combine Dataset (2017–2025)
3. Explore Data (EDA)
4. Handle Missing Values
5. Prepare Time Series (108 monthly observations)
6. Time Series Analysis (ADF Test, ACF/PACF)
7. Train-Test Split (84 / 24 months)
8. Model Training (ARIMA, SARIMA, Prophet)
9. Model Evaluation (MAE, MSE, RMSE, R^2)
10. Future Forecast – Next 6 Months (Jan–Jun 2026)
11. Conclusion & Insights